

BONDING – METALLIC BONDING

KEY VOCABULARY

Metallic bond: the strong attraction between positive metal ions and delocalised electrons.

Delocalised: electrons that are free to move through the whole structure.

Lattice: a regular, repeating arrangement of ions.

Malleable: can be hammered or bent into shape without breaking.

COMMON MISCONCEPTIONS

~~Metals are made of molecules.~~

✓ Metals are a giant lattice of ions.

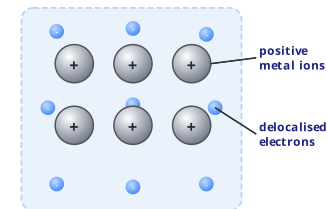
~~The electrons belong to one atom.~~

✓ Electrons are delocalised (shared by all).

~~Metals conduct because ions move.~~

✓ The delocalised electrons carry the charge.

THE METALLIC LATTICE



Positive **metal ions** sit in a 'sea' of **delocalised electrons**.

1 DEFINE THE BOND

Describe what is meant by **metallic bonding**. [2 marks]

Use the words 'positive metal ions' and 'delocalised electrons'.

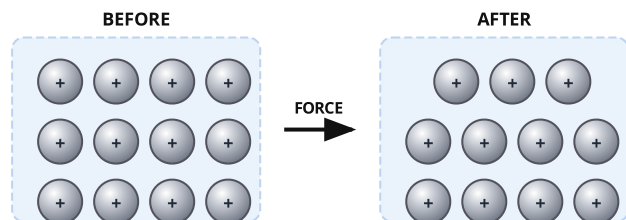
2 ELECTRICAL CONDUCTIVITY

Explain why metals are good conductors of electricity. [2 marks]

Think about which particles are free to move.

3 MALLEABILITY

The diagram shows the lattice before and after a force is applied. Explain why the layers of ions can slide. [2 marks]



4 MELTING POINT

Complete the sentence using the word bank. [3 marks]

Metals have ____ melting points because there is a ____ electrostatic ____ between the ions and the delocalised electrons.

high

strong

attraction

5 LABEL THE PARTICLES

In the lattice diagram, what do the large '+' circles and the small '-' circles represent? [2 marks]

6 WHY SO STRONG?

Suggest why a metal with ions of charge 2+ has a higher melting point than one with 1+ ions. [2 marks]
